

Electronic Supplementary Material:

Selective Support-Flow Candidate Generation with Cross-Slicer

Verification for Build-Orientation Search

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Status of this document. This supplement accompanies the prospective v2 manuscript. Geometry-only selection records, definitions, and verification procedures are fixed. Red fields remain intentionally empty until a hashed policy lock exists and the sealed external evaluation has been completed. The legacy v1 supplementary results are not imported into v2 outcome tables.

1 Reporting boundary and immutable identifiers

The existing 41-mesh G5 collection was used for method development and cannot serve as external validation. The v2 external manifest contains 60 meshes selected from 3,817 strict Thingi10K STL files without reading or computing a TOMO, Cura, or Prusa outcome. Thirty meshes were independently designated for the cross-slicer panel.

Table S1 Immutable v2 design identifiers at the geometry-selection stage.

Item	Value
Protocol ID	sftf-tdp-v2-2026-07-13
Manifest SHA256	e2671d0a257fd7b9437614ffc4f47df6ecfc79d15 e0b7177a7335e8d2e3eb797
External population	60 geometry-valid single-component watertight meshes
Cross-slicer population	30 meshes selected by a second salted hash
External-outcome state	Sealed; no external TOMO or slicer outcome computed
Policy-lock state	Not locked; all pending fields must be resolved first

2 Dimensionless SFTF v2 derivations

2.1 Normalized surface measure

Let μ_A be normalized surface-area measure on a triangular surface $\mathcal{S}$, so $\mu_A(\mathcal{S}) = 1$. A deterministic sample $\{(\mathbf{x}_k, \mathbf{m}_k)\}_{k=1}^K$ approximates expectation under μ_A and is reused for every build direction $\mathbf{n}$. Unlike the legacy direction-dependent source selection, the sampling distribution is therefore fixed when candidate directions are compared.

With $O_k = \max(0, -\mathbf{m}_k \cdot \mathbf{n})$, bounding-box diagonal D , receiver normal $\mathbf{r}_k$, and ray height h_k , the pair tensor is

$$\mathbf{F}_{\text{pair}}(\mathbf{n}) = \frac{1}{K} \sum_{k \in \mathcal{H}(\mathbf{n})} \frac{O_k}{1 + \lambda h_k / D} \mathbf{m}_k \mathbf{r}_k^T. \quad (\text{S1})$$

No receiver-triangle area multiplies this contribution. The scalar terms are

$$R(\mathbf{n}) = \max\{0, -\mathbf{n}^T \text{sym}(\mathbf{F}_{\text{pair}}) \mathbf{n}\}, \quad (\text{S2})$$

$$B(\mathbf{n}) = \frac{1}{K} \sum_{k \in \mathcal{P}(\mathbf{n})} O_k (1 + \lambda h_{k,\text{bed}} / D), \quad (\text{S3})$$

$$S_{\text{v2}}(\mathbf{n}) = R(\mathbf{n}) + B(\mathbf{n}). \quad (\text{S4})$$

2.2 Uniform-scale invariance

Under a uniform scale $s > 0$, $h \mapsto sh$, $h_{\text{bed}} \mapsto sh_{\text{bed}}$, and $D \mapsto sD$. Normals, overhang coefficients, and normalized surface measure are unchanged. Consequently,

$$\frac{sh}{sD} = \frac{h}{D}, \quad \frac{sh_{\text{bed}}}{sD} = \frac{h_{\text{bed}}}{D}, \quad (\text{S5})$$

and Eqs. (S1)–(S4) are invariant. This identity is checked both in Python on uniformly scaled meshes and symbolically in Mathematica 14.3.

2.3 Symmetric and antisymmetric parts

For $\mathbf{A} = -\mathbf{A}^T$ and any real $\mathbf{n}$,

$$\mathbf{n}^T \mathbf{A} \mathbf{n} = (\mathbf{n}^T \mathbf{A} \mathbf{n})^T = \mathbf{n}^T \mathbf{A}^T \mathbf{n} = -\mathbf{n}^T \mathbf{A} \mathbf{n} = 0. \quad (\text{S6})$$

Thus only $\text{sym}(\mathbf{F})$ can affect the Rayleigh scalar. The skew-to-symmetric Frobenius-norm ratio is eligible only as an uncertainty diagnostic.

2.4 Virtual build-plate identity

For a plate contribution $\mathbf{F}_{\text{bed},k} = (1 + \lambda h_{k,\text{bed}} / D) \mathbf{m}_k \mathbf{n}^T$, unit $\mathbf{n}$, and $\mathbf{n} \cdot \mathbf{m}_k = -O_k$,

$$-\mathbf{n}^T \text{sym}(\mathbf{F}_{\text{bed},k}) \mathbf{n} = O_k (1 + \lambda h_{k,\text{bed}} / D), \quad (\text{S7})$$

which is the scalar bed term used in the implementation.

3 Implementation and deterministic sampling

For each mesh, midpoint quadrature on cumulative triangle area selects source faces. Base-2 and base-3 radical-inverse coordinates place one deterministic point within each selected triangle. The

resulting points, source normals, face IDs, and D are cached in memory and reused for all candidate directions.

For each active overhang sample, a ray travels along $-\mathbf{n}$. All intersections are sorted by ray ID and distance. The first non-source face with positive source-to-hit height and receiver alignment $\mathbf{r}_k \cdot \mathbf{n} > 0$ is retained; otherwise the sample contributes to the virtual build plate. The default candidate score is lower-is-better.

Table S2 SFTF v2 implementation quantities to be frozen before external evaluation.

Quantity	Development values	Locked value
Surface samples K	2,048; 4,096; 8,192	Pending
Candidate pool	2,048 directions	Pending
Distance weight λ	1.0	Pending
Critical-angle gate	Coupon-calibrated	Pending
TOMO verification cells	Provisional 2,400	Pending
Top-basin expansion	Development-only comparison	Pending

4 External geometry-only selection

Candidate STL files were ranked by SHA256 of a fixed salt, relative path, and file SHA256. Known development IDs were excluded. Binary-STL size and face count were checked before loading. Selection then required finite vertices, positive surface area and bounding-box diagonal, watertightness, and one connected component. Exact duplicates were excluded by file SHA256.

Table S3 Face-count strata in the sealed external selection.

Face stratum	Eligible candidates	Examined	Holdout	Cura/Prusa
1k–10k	1,828	15	15	8
10k–50k	720	20	20	10
50k–150k	177	15	15	8
150k–250k	27	10	10	4
Total	2,752	60	60	30

All 60 examined in-stratum candidates passed geometry validation; the remaining 1,065 files were outside the predeclared face-count strata. No mesh was replaced based on an orientation outcome.

Table S4 Hash-locked 60-mesh external manifest.

Holdout	Thingy ID	Face stratum	Faces	Vertices	D (mesh units)	Cura/Prusa panel
H001	66485	10k–50k	12,156	6,076	94.874	Yes
H002	535202	10k–50k	12,486	6,245	88.474	Yes

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Table S4 (continued)

Holdout	Thingi ID	Face stratum	Faces	Vertices	D (mesh units)	Cura/Prusa panel
H003	486859	10k–50k	29,274	14,629	3.505	No
H004	153963	10k–50k	14,760	7,382	108.810	No
H005	112546	10k–50k	23,396	11,700	53.990	No
H006	55963	10k–50k	18,628	9,314	147.030	Yes
H007	108884	10k–50k	27,652	13,764	209.848	No
H008	103538	10k–50k	24,752	12,378	179.023	Yes
H009	579371	10k–50k	23,874	11,937	104.700	Yes
H010	1396893	10k–50k	20,256	10,128	35.305	No
H011	75133	10k–50k	30,792	15,398	121.528	No
H012	200082	10k–50k	41,008	20,506	45.525	Yes
H013	35269	10k–50k	40,246	20,125	136.017	Yes
H014	107910	10k–50k	25,990	12,997	51.919	Yes
H015	1312968	10k–50k	22,352	11,168	63.425	Yes
H016	190253	10k–50k	30,748	15,374	71.870	No
H017	925244	10k–50k	10,994	5,491	53.209	No
H018	106558	10k–50k	10,900	5,452	60.422	No
H019	301804	10k–50k	10,966	5,453	56.755	Yes
H020	69985	10k–50k	22,912	11,458	40.940	No
H021	353686	150k–250k	173,894	86,939	193.830	Yes
H022	601645	150k–250k	189,994	94,995	134.433	No
H023	1063863	150k–250k	168,258	84,129	95.469	No
H024	221686	150k–250k	182,690	91,347	152.125	Yes
H025	274379	150k–250k	191,204	95,604	151.245	Yes
H026	461109	150k–250k	156,064	78,034	71.434	No
H027	66480	150k–250k	181,504	90,754	151.826	No
H028	86164	150k–250k	188,984	94,494	114.479	Yes
H029	88053	150k–250k	153,840	76,920	257.780	No
H030	461111	150k–250k	153,248	76,626	88.412	No
H031	697213	1k–10k	4,966	2,481	32.135	Yes
H032	69986	1k–10k	1,428	716	42.264	Yes
H033	116896	1k–10k	2,104	1,054	39.744	No
H034	96653	1k–10k	1,152	566	167.275	No
H035	1512169	1k–10k	5,344	2,658	90.207	No
H036	103142	1k–10k	3,484	1,740	132.520	Yes
H037	723892	1k–10k	1,072	536	22.192	No
H038	124336	1k–10k	9,886	4,927	283.493	Yes
H039	1777452	1k–10k	1,766	885	105.577	No
H040	78738	1k–10k	6,726	3,365	67.393	No
H041	41521	1k–10k	4,372	2,188	150.000	Yes
H042	636795	1k–10k	7,636	3,820	71.431	No
H043	72669	1k–10k	2,266	1,131	100.455	Yes
H044	66108	1k–10k	2,056	1,024	79.361	Yes
H045	120791	1k–10k	1,048	524	240.244	Yes
H046	472052	50k–150k	132,308	66,152	247.229	Yes
H047	472037	50k–150k	61,884	30,942	61.315	No
H048	80557	50k–150k	59,256	29,626	143.582	Yes
H049	472205	50k–150k	82,358	41,179	47.790	No
H050	75652	50k–150k	66,414	33,209	54.337	Yes
H051	509316	50k–150k	57,600	28,564	196.151	No
H052	119829	50k–150k	126,756	63,366	143.488	Yes
H053	390936	50k–150k	95,136	47,570	63.340	No
H054	762603	50k–150k	94,994	47,499	23.326	No

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Table S4 (continued)

Holdout	Thingi ID	Face stratum	Faces	Vertices	D (mesh units)	Cura/Prusa panel
H055	395055	50k–150k	56,826	28,413	218.365	No
H056	472051	50k–150k	122,682	61,343	201.481	Yes
H057	270940	50k–150k	126,202	63,091	186.421	Yes
H058	762602	50k–150k	93,862	46,933	68.665	Yes
H059	96130	50k–150k	92,128	46,064	82.030	Yes
H060	399564	50k–150k	138,264	69,120	110.785	No

5 Policy lock and abstention rule

The policy lock is a machine-readable JSON file created after development-only analysis and before the first external outcome. It records the code commit, manifest hash, SFTF parameters, matched-budget construction, abstention feature definitions and thresholds, tie-breaking, statistical seeds and margin, the corrected TOMO binary hash, and both slicer profile hashes.

Table S5 Permitted uncertainty inputs and their required lock definitions.

Input	Definition required in the policy lock
Sampling stability	Angular/top-set agreement between fixed K values
Candidate crowding	Exact angular statistic over a fixed number of basins
Score dispersion	Exact normalization, quantile, and missing-value rule
Tensor skew ratio	Frobenius-norm expression and numerical floor
Policy decision	Boolean rule, thresholds, equality behavior, and tie-break

No external label, TOMO value, slicer output, semantic category, or filename token may enter the abstention rule.

6 Numerical verification and sensitivity plan

Table S6 Pre-performance numerical checks. Final tolerances are locked using development data only.

Check	Statistic	Locked tolerance
Uniform scaling	Relative score/rank change over fixed scale factors	Pending
Surface sampling	Top-set and score convergence for K values	Pending
Remeshing	Score/rank change after controlled subdivision/remesh	Pending
Rotation equivariance	Direction and score after rigid rotation	Pending
Python/compiled parity	Directional score and ranking agreement	Pending

The external performance experiment must not start if a locked numerical check fails. Failed checks and corrective development work are retained in the audit trail; thresholds cannot be relaxed after external outcomes are viewed.

7 Corrected TOMO provenance

The historical overflow occurred in per-slot support-volume accumulation. The v2 endpoints use a corrected 32-bit accumulator. A 16-bit legacy grid is not converted, clipped, or mixed with corrected results; it is excluded and retained only as an invalidated provenance artifact.

Table S7 Required TOMO provenance fields.

Field	Legacy record	v2 endpoint
Accumulator	int16, overflow affected	int32
Use	Invalidated provenance only	Primary intermediate verifier
DLL SHA256	Not accepted for v2	Pending
Source commit	Legacy record	Pending
Angle grid/support setting	Recorded if shown	Pending

8 Cura and Prusa calibration

Synthetic coupons with known downward surface slopes are evaluated before holdout meshes. Engine-specific support-angle settings are selected by a predeclared onset criterion; equal numeric settings are not assumed to mean equal physical thresholds. The exact configuration files are then hashed.

Table S8 Independent slicer executables and locked configuration fields.

Engine	Executable SHA256	Locked profile
CuraEngine 5.13	36E36D46...784AFDE2	Pending profile path, SHA256, angle, layer/nozzle/material/support settings
PrusaSlicer 2.9.6	7FD50B52...D9769704	Pending profile path, SHA256, angle, layer/nozzle/material/support settings
Legacy Cura 15.04.6	962E56D2...C612BA18	Continuity anchor only; not a primary endpoint

[**LOCKED RESULT TODO:** Insert coupon geometry, support-onset plots, parsed G-code checks, and the frozen setting decision.]

9 External corrected-TOMO results

The primary endpoint is grid-range-normalized regret (NRR), defined for all 60 meshes including zero-support references. The primary contrast is selective SFTF minus matched uniform, paired by mesh.

[**LOCKED RESULT TODO:** Insert the complete 60-row outcome table with reference range, uniform, SFTF-only, selective, abstention decision, random and legacy ablations, NRR, runtime, and failure flags. Generate the table from the immutable result JSON; do not hand-edit values.]

[**LOCKED RESULT TODO:** Insert bootstrap and permutation specifications, distributions, ECDF, 90th percentile, maximum regret, accepted coverage, and all predeclared sensitivity analyses.]

10 Independent Cura/Prusa results

Both modern slicers evaluate the identical locked direction union on the 30 preselected meshes. The best value within this finite union is described as a panel reference, not a global optimum. Cura and Prusa are analyzed separately and are not averaged into a synthetic truth.

[**LOCKED RESULT TODO:** Insert complete per-mesh Cura and Prusa support values, separate NRR contrasts, TOMO–slicer rank correlations, top-direction overlaps, disagreement cases, parse failures, and rerun audit.]

11 Runtime and reproducibility records

[**LOCKED RESULT TODO:** Insert hardware/OS/compiler details and separate wall times for surface sampling, SFTF candidate scoring, abstention, corrected-TOMO verification, Cura, Prusa, G-code parsing, and total pipeline execution. Report concurrency and both warm and cold cache policies.]

Every released result row must identify the protocol ID, manifest SHA256, policy-lock SHA256, source commit, mesh SHA256, backend binary/configuration hash, random seed where applicable, start/end timestamps, status, and error field. Machine-readable JSON/CSV files are the source of manuscript tables and figures.

12 Supplementary figure plan

The legacy TDP figures are retained only as source material. V2 figures are regenerated after policy lock using machine-readable results. Vector SVG is the archival source when available; raster-only images are stored as PNG.

Table S9 Planned v2 supplementary figures.

Figure	Content
S1	Uniform-scale, sampling-count, and remeshing sensitivity
S2	Overhang coupon geometry and engine-specific support onset
S3	All-mesh corrected-TOMO paired-regret distributions
S4	Abstention coverage-risk curve fixed from development data
S5	Cura and Prusa per-mesh agreement and disagreement panels
S6	Runtime decomposition and scaling with triangle count

13 Availability checklist

- Protocol, external manifest, manifest SHA256, and selection audit.
- Completed policy lock and SHA256.
- Corrected TOMO binary/source provenance.
- Cura and Prusa profiles, hashes, and coupon calibration outputs.
- Per-direction and per-mesh machine-readable outcome files.
- Scripts that regenerate every table and figure.
- Licensed retrieval instructions rather than redistribution of restricted mesh files.